Freight Intel AI Project Documentation

Simple User Guide & Architecture Documentation

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1. Project Overview (In Simple Words)

Freight Intel AI is an intelligent logistics application built to help businesses estimate shipment transit times, predict delivery delays, and prevent inventory stockouts before they happen.

Instead of relying on rough manual guesses, Freight Intel AI uses machine learning (XGBoost) trained on historical freight records to calculate exact delay risks, estimated financial penalty costs in Indian Rupees (■), and actionable stock cushion recommendations.

2. Key Features & Capabilities

- **Smart Machine Learning Delay Predictor:** Calculates exact delay margins (in days), overall transit duration, penalty costs, and classifies risk into **Low, Moderate, or High Risk** tiers.
- **Interactive 3D Supply Chain Globe:** Built with Three.js. Spin, zoom, and inspect real-time animated comet trails mapping shipping routes across Indian logistics hubs (Mumbai, Delhi, Chennai, Kolkata, Bengaluru, Hyderabad, Ahmedabad, Kochi).
- **Live Destination Weather Integration:** Fetches live weather conditions automatically from Open-Meteo API for real-time weather risk evaluation.
- **Analytics Dashboard & Retraining:** Visualizes carrier distributions, transport mode metrics, and feature importance. Allows single-click model retraining on updated logs.
- **Export PDF Risk Reports:** Instantly generates downloadable risk summary reports for freight managers.

3. Design & Theme System (Shopify Editions Style)

The web application features an elegant **Periwinkle & Lavender Light Theme** designed for modern aesthetics and high text readability:

Hex Code	Color Name	UI Application
#3D52A0	Deep Periwinkle	Primary titles, CTA buttons, metrics headers, dark periwinkle text.
#7091E6	Soft Cornflower Blue	Secondary buttons, active route highlights, glowing comets.
#8697C4	Steel Periwinkle	Subtle borders, 3D stars, muted text labels.
#ADBBDA	Ice Lavender	Glass card borders, input borders, diagnostic containers.
#EDE8F5	Soft Lavender Off-White	Page canvas background & 3D background fog.

4. Quick Start Guide (How to Run)

Follow these steps to run Freight Intel AI locally:

1. Open your terminal in the project directory: d:\Projects\Freight Inventory Analysis
2. Activate the virtual environment: .\venv\Scripts\Activate.ps1
3. Train or update machine learning model weights: python train_model.py
4. Launch the application server: python app.py (or .\venv\Scripts\python.exe app.py)
5. Open your web browser at: **http://127.0.0.1:5000**

5. Technical Stack Summary

Layer	Technologies Used
Backend Core	Python 3.10+, Flask Web Server, Joblib Serialization
Machine Learning	XGBoost Regressor, Scikit-learn, Pandas, NumPy
Frontend UI & 3D	HTML5, Vanilla CSS3 (Glassmorphism), Three.js (WebGL), OrbitControls, Bootstrap 5
Data & Persist	CSV Raw Log Persister (prediction_history.csv), Open-Meteo Weather API